

Nachiket Mahajan

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EDUCATION

Bachelor of Technology in Electronics Engineering (MIT AOE), Pune

(2023 – Present)

Specialization: Embedded Systems, PCB Design, Machine Learning, Computer vision

• CGPA (Third year): 9.09 | SGPA (TY 6th sem): 9.29

10th | SSc | D.Y. Patil English medium School | 84.20%

Jan 2009 - April 2021

12th | HSc | D.Y Patil, Shahunagar | 73.83%

Jan 2021 - April 2023

WORK EXPERIENCE

Research Summer Intern | 3D Computer vision

May 2026 - Present

Clarity Lab, IIT jodhpur

on site, Jodhpur

• Designed a gesture-driven 3D scene navigation and object manipulation system using hand tracking, ray casting, and real-time gesture recognition.

• Implemented interactive 3D rendering and scene exploration tools using Ursina Engine for visualization of large-scale spatial structures.

• Developed Mixed Reality prototypes on Meta Quest featuring hand-tracked interaction with virtual objects and immersive spatial experiences. .

R&D Intern | Embedded Systems Engineer

Jun 2025 - May 2026

Inthink Technologies

Hybrid, Pune

• Designed and developed an **embedded healthcare system** for non-invasive diabetes monitoring.

• Worked with STM32 microcontrollers (STM32L432KC, STM32WB55CGU6), embedded C firmware development and communication protocols including UART, I²C, RS-485, and BLE.

Data Automation Intern

Aug 2024- Jan 2025

Zumax Innovation

Remote, Pune

• Developed a **Python-based** automation pipeline to process steel material data (thickness, length, weight, type) from CSV files, eliminating manual data entry and automatically logging date, time, and complete records.

• Built a web-based interface using Flask with HTML, CSS, JavaScript to manage and visualize the data.

COURSES & CERTIFICATIONS

• **Digital Twin** - MITAOE | ISA Pune Section | IEEE Bombay Section - [Certificate](#)

• **IoT Architectural Engineering** - L&T - [Certificate](#)

Projects

• **Applied Machine learning techniques for the classification of schizophrenia based on features extracted from multi-channel EEG signals:**

Major Project – under the Guidance of Dr. Vipin P. Yadav

Implemented ML algorithms for region-wise EEG signals analysis using Double Banana Montage and extracted spectral, statistical, and entropy-based features from multi-channel EEG signals for schizophrenia classification. Comparative evaluation across multiple ML models and feature selection methods demonstrated consistently high classification performance across frontal, temporal, parietal, and occipital brain regions, accuracies ranging from 86.84% to 97.74%

• **An Integrated Vision and Geolocation mapping System for Sand Grain Size Analysis:** [link](#)

Designed and implemented a vision-based system Using Raspberry Pi for camera control and GPS-based location tagging, with image processing algorithms like Watershed and segmentation for automated grain detection and dimensional analysis and a Flask-based web interface to visualize the locations, store, and manage analysis results.

• **Real-Time Vision-Based Hand Gesture Recognition for Multi-Degree-of-Freedom Robotic arm Control:**

under the Guidance of Dr. Abhilasha Joshi

Designed and implemented a hardware-integrated, real-time hand gesture recognition system for controlling a multi-DOF robotic arm. The system used computer vision and machine learning to extract and classify hand landmarks via MediaPipe, and interfaced the trained model with a servo-driven robotic arm through serial communication for real-time motion control

• **Vision-Based Numerical Extraction from Multi Seven-Segment Displays in CNC Machines using C++:**

Implemented a complete image processing pipeline in C++ including grayscale conversion, adaptive thresholding, noise trimming, digit segmentation, and seven-segment pattern decoding. Optimized algorithms were designed to accurately detect and interpret digits arranged in a single row (e.g., 123, 456, 4789) while minimizing computational complexity and hardware requirements, making the system suitable for real-time industrial monitoring applications.

TECHNICAL SKILLS

• **Programming Languages:** Python (Libraries: NumPy, Pandas, Matplotlib, Paho(mqtt), Flask, opency, sklearn, tensorflow), C++, Embedded C, HTML, CSS.

• **Computer Vision & Image Processing:** image segmentation, morphological operations, feature extraction, object detection, and geometric measurement from 2D images.

• **PCB Design & Prototyping:** practical knowledge of circuit simulation and design tools including Multisim, Proteus, KiCad.

• **Communication protocols:** I²C, SPI, UART/USART, RS-485 (Modbus RTU), Bluetooth Low Energy (BLE), Serial Communication, HTTP/REST

• **Embedded Software & Microcontrollers:** Skilled in using STM32CubeIDE, STM32CubeProgrammer, STM32CubeStudio, STM32CubeMonitor and Arduino IDE for embedded system development. Familiar with STM32F4, STM32N6, STM32L4, STM32WB series, Rasbery pi, esp32/esp8266, 8051, Arduino microcontrollers.

• Efficient in making **attractive & interactive PPT**, good in excel and word also proficient in using latex

SOFT SKILLS

• Effective Communication, Project Planning, Team Collaboration & Presentation.

• Creative skills: Crochet, knitting, Sketching and Painting, Cooking and Baking.